

Response to Reviewers

ICIC–APTIKOM 2026, Paper ID 57

Benchmarking Multi-View Tree-Level Oil Palm Bunch Counting Under a Fixed Detector

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Revised manuscript Camera-ready version, 6 pages, certified by IEEE PDF eXpress (file ID 2026332983)

General Response

We thank the four reviewers for their careful reading and constructive comments. We addressed every comment in the revised manuscript. The revision keeps the original experimental design and the six-page limit, and adds the following analyses:

- Tree-level bootstrap 95% confidence intervals (CIs), exact paired permutation tests and Holm correction for all counter and feature comparisons (Sections II-E and III-A to III-C).
- A second detector family (YOLO11m) trained on the official split, four YOLO26 checkpoints and a confidence-threshold sweep (Sections II-B and III-E, Table VI).
- Appearance-level matching that separates missed annotations, wrong-class detections and false positives, with diagnostic counting rules (Table I, Section III-D, Fig. 2).
- Per-class MAE, RMSE, signed bias and total-count error (Section II-D, Tables III and V).
- Validation-based model selection, matched repeated cross-validation (CV) and cross-estate counter transfer (Sections II-E, III-C and III-E).
- A feature-definition table (Table II), an analysis of the Random Forest (RF) results (Section III-F) and a limitations subsection (Section III-G).

During the re-analysis we also found and corrected three errors in the submitted version; they are listed in Response 3.10.

Section, table and figure numbers below refer to the revised manuscript. The submitted Tables II and III are merged into the revised Table III, the submitted Table V is now Table IV, and the submitted Table VI is now Table V. Code, cached predictions and the outputs of all added analyses are available at <https://github.com/ULM-SawitMVC/Baseline-SawitMVC> (folders `experiments/revision` and `results/revision`). The abbreviation pp denotes percentage points, and p_H denotes a Holm-adjusted p -value.

Reviewer 1

CONCLUSION: Include further implications for the field. Discuss benefits or shortcomings of your work and suggest future areas for research.

Response. We thank the reviewer for the positive assessment. We rewrote Section IV to cover the three requested points.

- *Implications.* Evaluating the same counter under ground-truth (GT) and detector inputs lets practitioners separate the effect of input quality from the effect of aggregation. For the Black Bunch Census, the results support improving B3 and B4 detection and B2-to-B3 classification together with aggregation. Class-level and total-count metrics should both be reported, because a plausible total count can hide class errors that change the harvest schedule.
- *Benefits.* Cached predictions, fixed splits and released code allow new counters to be compared without repeating detector inference.
- *Shortcomings.* The evidence covers one dataset from two estates of unequal size, and acquisition changes are untested. The results do not give an association-error budget or a limit on what future counters can achieve. Section III-G lists the remaining limitations.
- *Future research.* Comparison of explicit multi-view association with learned aggregation under matched inputs, broader detector evaluation, repeated detector training, and tests on an independent plantation and at measured camera-pitch angles.

Changes in manuscript: Section IV (Conclusion); Section III-G (Limitations).

Reviewer 2

However, the current manuscript does not yet provide sufficient evidence to support some of its broader conclusions. The use of a single detector, limited test set, absence of statistical significance analysis, and limited error decomposition are the main weaknesses. I would recommend acceptance after minor revision, provided that the authors strengthen the statistical analysis, clarify the scope of their conclusions, and provide a more rigorous decomposition of detector and counting errors.

Response. We thank the reviewer for recognising the value of the controlled comparison. We addressed the four weaknesses as follows. The responses to Reviewer 3 give further details.

- (a) *Single detector.* We trained a second detector family, YOLO11m, on the official 716/96/141 split with the same training settings. With the same Ridge+ F_0 counter on the 141 test trees, Class ± 1 accuracy is 73.76% for YOLO11m, 76.06% for YOLO26m and 97.70% for GT inputs. GT inputs remain 23.94 pp better than YOLO11m (95% CI 20.57–27.30; $p < 0.001$), while the two detectors do not differ significantly (-2.30 pp; CI -4.96 to $+0.35$; $p = 0.118$). Four YOLO26 checkpoints span 3.13 pp, compared with a 23.44 pp gap to GT inputs (Response 3.2).
- (b) *Limited test set.* Matched repeated CV uses identical folds under both input conditions. On 237 trees outside detector gradient training, Class ± 1 is $97.72 \pm 1.02\%$ (GT) versus $72.70 \pm 2.35\%$ (fixed detector); on the 141 original test trees it is $97.16 \pm 1.23\%$ versus $74.08 \pm 2.96\%$. A validation-only model selection gives a gap of 23.58 pp (Responses 3.1 and 3.5).
- (c) *Significance analysis.* The class-level gap between the best configurations is 20.57 pp (CI 17.20–23.94; $p < 0.001$). With the same Ridge+ F_0 counter in both conditions, it is 21.63 pp (CI 18.44–25.00; $p < 0.001$), so the gap is not caused by comparing different counter families. No pair of counters differs significantly under the fixed detector after Holm correction (Response 3.1).
- (d) *Error decomposition.* Appearance-level matching reports missed, wrong-class and false-positive counts for each class, and 22.7% of test bunches are not detected in any view. Diagnostic counting rules show the effect of each error type (Responses 3.3 and 3.9).

We also narrowed the conclusions to the tested configurations on SawitMVC (Response 3.8).

Changes in manuscript: Abstract; Sections II-B, II-E, III-A to III-E and IV; Tables I and VI; Fig. 2.

Reviewer 3

Comment 3.1

Add statistical uncertainty/significance analysis for the reported model differences.

Response. Section II-E defines the protocol. Percentile bootstrap CIs resample trees 10,000 times (seed 42), keep the four class counts of each tree together, and reuse the same resamples for paired comparisons. Exact paired permutation tests exchange the outcomes of two models within each tree. Holm correction covers the ten counter pairs within each input condition and, as a separate family, the seven feature sets compared with F_0 . Wilcoxon tests on per-tree macro absolute error and exact McNemar tests on Tree ± 1 cover the other endpoints. The main results are:

- GT inputs: the four counters other than RF do not differ significantly ($p_H = 1$). ElasticNet exceeds RF by 2.13 pp (CI 0.89–3.55; $p = 0.0042$, $p_H = 0.0376$), and SVM also exceeds RF ($p_H = 0.0342$).
- Fixed detector: none of the ten counter pairs differs significantly. The largest difference, ElasticNet versus RF, is 3.19 pp ($p = 0.0628$, $p_H = 0.628$).
- Input conditions: 20.57 pp (CI 17.20–23.94) between the best configurations and 21.63 pp (CI 18.44–25.00) for the same counter; both $p < 0.001$.

The manuscript states that a nonsignificant difference does not establish equivalence. Because the submitted version highlighted the configuration with the best test score, we added a validation-only selection over 40 model and feature candidates per condition. The selected configurations, Ridge+ F_0 (GT) and SVM+(F_0 + conf) (fixed), reach 97.70% and 74.11% on the test trees, a gap of 23.58 pp (CI 20.21–27.13; $p < 0.001$).

Changes in manuscript: Sections II-E, III-A, III-B and III-C.

Comment 3.2

Clarify the scope of the detector-related conclusion, because only one detector checkpoint is evaluated.

Response. We agree that one checkpoint cannot support a general statement. We added three analyses (Table VI) and restricted the conclusion accordingly.

- *Detector family.* YOLO11m was trained on the official split with 60 epochs, batch 32, image size 640, patience 60 and seed 42; its checkpoint was selected on the validation trees. The results are given in Response 2(a). YOLO11m misses 976 and misclassifies 399 of the 2,612 GT appearances.
- *Checkpoints.* YOLO26n, YOLO26s and YOLO26m from an earlier split, and YOLO26m from the official split, reach 70.70–73.83% Class ± 1 on 64 trees outside the gradient training of all checkpoints, compared with 97.27% for GT inputs.
- *Operating point.* Over eleven confidence thresholds from 0.05 to 0.70, with the counter refitted at each threshold, no threshold exceeds 77.48% Class ± 1 . Threshold 0.10 raises Tree ± 1 from 32.62% to 38.30%.

Section III-G states the remaining limits: one training run per detector, different Ultralytics versions and defaults (8.4.49 and 8.4.140), and confounding of model capacity with training split in the checkpoint analysis. The manuscript therefore describes the finding as applying to the tested detectors on SawitMVC, not as a general ranking of architectures.

Changes in manuscript: Sections II-B, III-E and III-G; Table VI.

Comment 3.3

Provide more detailed error decomposition for missed detections, false positives, and maturity-class confusion.

Response. Section III-D matches every detection on the test trees to the annotations. The matching is greedy in order of confidence, class-agnostic, and accepts a pair at $\text{IoU} \geq 0.5$. Table I reports, for each class, annotations with no matched detection (Missed), matched boxes with an incorrect class (Wrong) and detections with no matched annotation (FP).

- Of 2,612 annotated appearances, 964 (36.9%) are missed, 396 (15.2%) have an incorrect class and 1,252 (47.9%) are correct.
- Of 2,354 detections, 706 (30.0%) are false positives. Their mean confidence is 0.337, compared with 0.487 for matched detections.
- B2 errors are mainly class confusion (209 appearances, of which 158 are predicted as B3, versus 161 missed). B4 errors are mainly missed detections (262 of 455).
- Of 1,397 physical test bunches, 317 (22.7%) are not detected in any view, from 8.5% for B1 to 42.3% for B4.

The manuscript notes that an unmatched annotation can also result from an IoU below the threshold or from matching competition.

Changes in manuscript: Table I; Section III-D.

Comment 3.4

Add additional counting metrics, especially per-class MAE/RMSE and total bunch-count error.

Response. Section II-D defines macro MAE, macro RMSE, signed class bias, total-count MAE and total ± 1 accuracy. Table III reports macro MAE for all counters under both input conditions, and Table V gives per-class accuracy, MAE, RMSE and bias for the best configuration in each condition. From GT to fixed-detector inputs, macro RMSE rises from 0.522 to 1.418, total-count MAE rises from 0.979 to 1.787, and total ± 1 accuracy falls from 78.01% to 48.94% (Section III-C). The manuscript distinguishes total ± 1 , in which class errors can cancel, from Tree ± 1 , which requires all four class counts to be within one bunch.

Changes in manuscript: Sections II-D and III-C; Tables III and V.

Comment 3.5

Strengthen validation using repeated splits, cross-validation, or an independent plantation if available.

Response. No independent plantation was available for this revision. We therefore added three resampling analyses and state the missing independent test as a limitation.

- *Matched repeated CV.* Five folds with five repeats, stratified by estate and dominant class, use identical folds for GT and fixed-detector inputs with the detector frozen. Class ± 1 is $97.72 \pm 1.02\%$ versus $72.70 \pm 2.35\%$ on 237 trees outside detector gradient training (gap 25.03 pp), and $97.16 \pm 1.23\%$ versus $74.08 \pm 2.96\%$ on the 141 original test trees (gap 23.08 pp).
- *Validation-based selection.* Model and features are selected on the validation trees only; the test gap is 23.58 pp (Response 3.1).
- *Cross-estate transfer.* Ridge+ F_{all} trained on 641 DAMIMAS trees reaches 71.88% on 24 LONSUM evaluation trees, compared with 72.92% when trained on 75 LONSUM trees. The reverse transfer reaches 68.78% on 213

DAMIMAS trees. Because the detector was trained on both estates, this analysis evaluates the counter only.

Changes in manuscript: Sections II-E, III-C, III-E and III-G.

Comment 3.6

Expand the feature-definition table to improve reproducibility.

Response. Table II now defines all 67 dimensions: F_0 (13), confidence (20), spatial (8), distribution (20) and composition (6). It gives the formulas, the normalisation of box position and area by image size, the confidence thresholds ($p \geq 0.5$ and $p \geq 0.6$), the population standard deviation over all views including empty views, $\varepsilon = 10^{-6}$, and the default values for a class with no detection. Section II-C defines the notation, and the released feature extractor produces the same 67 dimensions. We also report that the 1.42 pp gain of F_{all} over F_0 is not statistically resolved (CI -0.53 to $+3.37$; $p = 0.201$, $p_H = 1$) and that F_{all} does not lead under repeated CV (Table IV).

Changes in manuscript: Section II-C; Tables II and IV.

Comment 3.7

Discuss the poor Random Forest performance more thoroughly.

Response. Section III-F explains the result. Each regression tree averages the training targets within a leaf, so RF cannot predict beyond the training count range and pulls estimates towards common counts when high counts are rare. The diagnostics agree with this explanation:

- Slopes of predicted on true counts are 0.754–0.888 for RF and 0.865–0.951 for the linear counters, and the relative prediction standard deviation of RF is smaller in every class.
- RF never predicts more than three B1 bunches, although the maximum is five in training and six in the test set; its B1 MAE is 0.149, compared with 0.050 for Ridge.
- In the highest-load bin, RF trails ElasticNet by 3.07 pp (94.74% versus 97.81%).

Under GT inputs, RF is significantly worse than ElasticNet and SVM after Holm correction; under the fixed detector the difference is not resolved ($p = 0.0628$, $p_H = 0.628$). The manuscript presents these findings as diagnostics and does not exclude improvement by tuning.

Changes in manuscript: Section III-F.

Comment 3.8

Moderate broad generalizations about detector quality versus counter quality.

Response. We revised the following statements:

- The submitted abstract stated that “the dominant finding is that detector output quality, more than the choice of counter, limits deployed accuracy.” The revised abstract states that the results “support prioritising detector output quality on this dataset” and “do not establish a ceiling on counter improvement.”
- The submitted statement that the remaining counting problem is “simple enough for compact regression models” was removed.
- GT inputs are described as an empirical reference, not a mathematical upper bound on future counting methods (Section II-B).
- Nonsignificant differences are not interpreted as equivalence (Sections II-E, III-A and III-E).
- The conclusion is limited to the tested configurations and lists the evidence needed before the claims can be extended to operational harvest forecasts (Section IV).

Changes in manuscript: Abstract; Sections II-B, II-E, III-A to III-G and IV.

Comment 3.9

Clearly distinguish detector performance limitations from multi-view association/aggregation limitations.

Response. Section III-D and Fig. 2(a) compare six counting rules on the 141 test trees:

Counting rule	Class ± 1 (%)
GT boxes and classes, ElasticNet+ F_0	98.05
Matched physical bunches counted once, GT class	86.17
Same identities, confidence-weighted detector class vote	71.81
Class vote, each unmatched detection counted as a separate bunch	62.23
Deployed counter, Ridge+ F_{all}	77.48
Naive sum of detections	50.71

The three identity-based rules show the separate effects of bunches that are never detected, class confusion and unmatched detections. The manuscript states that these rules are separate estimators and that their differences do not add up to the deployed error. The deployed counter (77.48%) exceeds two of the rules because learned aggregation compensates statistically for missed counts and class bias. A direct measurement of association error requires an explicit association module, which we list as future work.

Changes in manuscript: Section III-D; Fig. 2(a); Section IV.

Comment 3.10

Carefully proofread the manuscript for formatting, equations, tables, and citation consistency.

Response. We proofread the full manuscript. All tables were re-typeset (Response 4.1), equations (1) and (2) define Class ± 1 and Tree ± 1 , abbreviations are defined at first use, and references are numbered in order of first citation. The final PDF passed IEEE PDF eXpress. During the re-analysis we found and corrected the following errors in the submitted version:

- *Global divisor (GT inputs).* Class ± 1 / Tree ± 1 / MAE were corrected from 95.39% / 85.11% / 0.376 to 95.57% / 86.52% / 0.356. The divisor is fitted on the 716 training trees only, separately for each input condition ($k = 1.891$ for GT and 1.793 for the fixed detector). A fixed-detector row (70.21% / 22.70% / 1.188) was added to Table III.
- *Tree ± 1 gap.* The value was corrected from 59.58 to 59.57 pp.
- *Visibility-adaptive divisor.* This row was removed because we could not confirm that the trees used to develop the heuristic are disjoint from the current test split.
- *Software citation.* The unsupported version number “26.0.0” was removed from the Ultralytics reference; the training log records Ultralytics 8.4.49.

Changes in manuscript: All sections; Tables I to VI; references.

Reviewer 4

We thank the reviewer for the positive assessment of the problem statement, dataset and methodology.

Comment 4.1

Fix several noticeable typesetting and column-alignment errors in Tables III and V.

Response. The submitted Table III (fixed-detector counting) is now merged with the submitted Table II into the revised Table III, which places GT and fixed-detector results side by side. The submitted Table V (feature ablation) is now Table IV. All six tables were re-typeset with 8-point text, horizontal rules only, the same number of decimal places in each column, and widths within the column. Captions were shortened, and units and definitions were moved to table notes.

Changes in manuscript: Tables I to VI.

Comment 4.2

Briefly evaluate an additional detector architecture to strengthen the generalizability of the findings.

Response. We trained YOLO11m on the official 716/96/141 split with the same epochs, batch size, image size, patience and seed as the YOLO26m detector, and selected its checkpoint on the validation trees. With the same Ridge+ F_0 counter on the 141 test trees:

Input	Detections	Class ± 1 (%)	Tree ± 1 (%)
YOLO11m	2,099	73.76	26.24
YOLO26m	2,354	76.06	28.37
GT detections	2,612	97.70	90.78

The two detectors do not differ significantly in Class ± 1 (-2.30 pp; CI -4.96 to $+0.35$; $p = 0.118$), and GT inputs remain 23.94 pp better than YOLO11m (CI 20.57 – 27.30 ; $p < 0.001$). The manuscript notes that one training run per detector and different software versions limit wider generalisation.

Changes in manuscript: Sections II-B, III-E and III-G; Table VI.

Comment 4.3

Clarify how robust the vertical spatial features are to real-world camera pitch variations.

Response. The dataset has no calibrated pitch measurements and no repeated captures at known angles, so physical pitch robustness could not be tested. We added a sensitivity analysis of the vertical feature. The mean vertical position of the test features was transformed as $\overline{cy} \mapsto \text{clip}(a \overline{cy} + b, 0, 1)$ over 35 settings of $a \in [0.90, 1.10]$ and $b \in [-0.10, 0.10]$, with training data unchanged and missing-class values kept at 0.5. Ridge+ F_{all} loses at most 1.42 pp Class ± 1 , and Gaussian noise with $\sigma = 0.05$ costs 0.53 pp. This analysis tests feature shifts only. A real pitch change also affects perspective, box area, visibility and detector output. The manuscript therefore states that robustness to camera rotation remains unverified (Sections III-E and III-G) and proposes repeated captures at measured pitch angles as future work (Section IV).

Changes in manuscript: Sections III-E, III-G and IV.